AMS 526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 16: QR Algorithm without Shifts (NLA §28); QR Algorithm with Shifts (NLA §29)

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Outline

1. **QR Algorithm without Shifts**
2. QR Algorithm with Shifts

QR Algorithm

- The most basic version of the QR algorithm is remarkably simple.

"Pure" QR Algorithm

$A^{(0)} = A$

for $k = 1, 2, \ldots$

$$Q^{(k)} R^{(k)} = A^{(k-1)}$$
$$A^{(k)} = R^{(k)} Q^{(k)}$$

- With some suitable assumptions, $A^{(k)}$ converges to the Schur form of A (diagonal if A is symmetric).
- The similarity transformation of A is given by

$$A^{(k)} = R^{(k)} Q^{(k)} = \left(Q^{(k)}\right)^{\mathsf{T}} A^{(k-1)} Q^{(k)}.$$

- But why does it work?

Unnormalized Simultaneous Iteration

- To understand the QR algorithm, first consider a simple algorithm.
- Simultaneous iteration is power iteration applied to several vectors.
- Start with linearly independent $v_1^{(0)}, \dots, v_n^{(0)}$.
- We know from power iteration that $A^k v_1^{(0)}$ converges to q_1.
- With some assumptions, the space $\left\langle A^k v_1^{(0)}, \dots, A^k v_n^{(0)} \right\rangle$ should converge to $\langle q_1, \dots, q_n \rangle$.
- Define the initial matrix $V^{(0)}$ and matrix $V^{(k)}$ at step k as

$$V^{(0)} = \left[v_1^{(0)} \mid \cdots \mid v_n^{(0)} \right], \; V^{(k)} = A^k V^{(0)} = \left[v_1^{(k)} \mid \cdots \mid v_n^{(k)} \right].$$

Unnormalized Simultaneous Iteration

- Define an orthogonal basis for the column space of $V^{(k)}$ by the QR factorization $\hat{Q}^{(k)}\hat{R}^{(k)} = V^{(k)}$.
- We assume that
  1. the leading $k+1$ eigenvalues are distinct, and
  2. all leading principal submatrices of $\hat{Q}^{\mathsf{T}}V^{(0)}$ are nonsingular, where $\hat{Q} = [q_1 \mid \cdots \mid q_n]$.
- We then have the columns of $\hat{Q}^{(k)}$ converge to the eigenvectors of A, such that

$$\|q_j^{(k)} - (\pm q_j)\| = \mathcal{O}(c^k),$$

  where $c = \max_{1\le i\le n-1} |\lambda_{i+1}|/|\lambda_i|$.
- Proof idea: Show that the subspace of any leading j columns of $V^{(k)} = A^k V^{(0)}$ converges to the subspace of the first j eigenvectors of A, so does the subspace of any leading j columns of $\hat{Q}^{(k)}$.

Simultaneous Iteration

- The matrices $V^{(k)} = A^k V^{(0)}$ are highly ill-conditioned.
- We orthonormalize at each step rather than at the end.

Simultaneous Iteration

Pick $\hat{Q}^{(0)} \in \mathbb{R}^{n \times n}$

for $k = 1, 2, \ldots$

$$Z = A\hat{Q}^{(k-1)}$$

$$\hat{Q}^{(k)}\hat{R}^{(k)} = Z$$

- The column spaces of $\hat{Q}^{(k)}$ and $Z^{(k)}$ are both equal to the column space of $A^k\hat{Q}^{(0)}$, therefore achieving the same convergence as before.

Simultaneous Iteration with $\hat{Q}^{(0)} = I \iff$ QR Algorithm

Simultaneous Iteration

Pick $\hat{Q}^{(0)} \in \mathbb{R}^{n\times n}$

for $k = 1, 2, \ldots$

$$Z = A\hat{Q}^{(k-1)}$$
$$\hat{Q}^{(k)}\hat{R}^{(k)} = Z$$

"Pure" QR Algorithm

$A^{(0)} = A$

for $k = 1, 2, \ldots$

$$Q^{(k)}R^{(k)} = A^{(k-1)}$$
$$A^{(k)} = R^{(k)}Q^{(k)}$$

- The QR algorithm is equivalent to simultaneous iteration with $\hat{Q}^{(0)} = I$.
- Replace $\hat{R}^{(k)}$ by $R^{(k)}$ and $\hat{Q}^{(k)}$ by $\underline{Q}^{(k)}$, and introduce the new statement $A^{(k)} = \left(\underline{Q}^{(k)}\right)^{\mathsf{T}} A\underline{Q}^{(k)}$ in simultaneous iteration.

Simultaneous iteration

$$\underline{Q}^{(0)} = I$$
$$Z = A\underline{Q}^{(k-1)}$$
$$\underline{Q}^{(k)}R^{(k)} = Z$$
$$A^{(k)} = \left(\underline{Q}^{(k)}\right)^{\mathsf{T}} A\underline{Q}^{(k)}$$

QR algorithm

$$A^{(0)} = A$$
$$Q^{(k)}R^{(k)} = A^{(k-1)}$$
$$A^{(k)} = R^{(k)}Q^{(k)}$$
$$\underline{Q}^{(k)} = Q^{(1)}Q^{(2)}\cdots Q^{(k)}$$

Simultaneous Iteration with $\hat{Q}^{(0)} = I \iff$ QR Algorithm

- $\underline{Q}^{(k)} = Q^{(1)}Q^{(2)}\cdots Q^{(k)}$. Let $\underline{R}^{(k)} = R^{(k)}R^{(k-1)}\cdots R^{(1)}$.
- Both schemes generate the QR factorization $A^k = \underline{Q}^{(k)}\underline{R}^{(k)}$ and projection $A^{(k)} = \left(\underline{Q}^{(k)}\right)^{\mathsf{T}} A\underline{Q}^{(k)}$.

We prove this by induction. For $k = 0$, it is trivial for both algorithms.
For $k \geq 1$ with simultaneous iteration, $A^{(k)}$ is given by definition, and we have

$$A^k = A\underline{Q}^{(k-1)}\underline{R}^{(k-1)} = \underline{Q}^{(k)}R^{(k)}\underline{R}^{(k-1)} = \underline{Q}^{(k)}\underline{R}^{(k)}.$$

For $k \geq 1$ with the QR algorithm, we have

$$A^k = A\underline{Q}^{(k-1)}\underline{R}^{(k-1)} = \underline{Q}^{(k-1)}A^{(k-1)}\underline{R}^{(k-1)} = \underline{Q}^{(k)}\underline{R}^{(k)},$$

and

$$A^{(k)} = \left(Q^{(k)}\right)^{\mathsf{T}} A^{(k-1)}Q^{(k)} = \left(\underline{Q}^{(k)}\right)^{\mathsf{T}} A\underline{Q}^{(k)}.$$

Convergence of the QR Algorithm

- Since $\underline{Q}^{(k)} = \hat{Q}^{(k)}$ in simultaneous iteration, the column vectors of $\underline{Q}^{(k)}$ converge linearly to the eigenvectors if A has distinct eigenvalues.
- From $A^{(k)} = \left(\underline{Q}^{(k)}\right)^{\mathsf{T}} A \underline{Q}^{(k)}$, the diagonal entries of $A^{(k)}$ are Rayleigh quotients of the column vectors of $\underline{Q}^{(k)}$, so they converge linearly to the eigenvalues of A.
- The off-diagonal entries of $A^{(k)}$ converge to zeros, as they are generalized Rayleigh quotients involving approximations of distinct eigenvectors.
- Overall, $A = \underline{Q}^{(k)} A^{(k)} \left(\underline{Q}^{(k)}\right)^{\mathsf{T}}$. For a symmetric matrix, it converges to the eigenvalue decomposition of A.
- The convergence rate is only linear: the columns of $\hat{Q}^{(k)}$ converge to the eigenvectors of A such that $\|q_j^{(k)} - (\pm q_j)\| = \mathcal{O}(c^k)$, where $c = \max_{1 \le i \le n-1} |\lambda_{i+1}|/|\lambda_i|$.

Outline

1. QR Algorithm without Shifts
2. **QR Algorithm with Shifts**

Simultaneous Inverse Iteration $\Longleftrightarrow$ QR Algorithm

- Similar to inverse iteration, the QR algorithm can be sped up by introducing shifts at each step.
- Assume A is real and symmetric. The QR algorithm is equivalent to simultaneous inverse iteration, applied to the "flipped" identity matrix P given by

$$P = \begin{bmatrix} & & & 1 \\ & & 1 & \\ & \vdots & & \\ 1 & & & \end{bmatrix}.$$

Simultaneous inverse iteration

$\hat{Q}^{(0)} = P$

for $k = 1, 2, \ldots$

$\quad Z = A^{-1}\hat{Q}^{(k-1)}$

$\quad \hat{Q}^{(k)}\hat{R}^{(k)} = Z$

"Pure" QR Algorithm

$A^{(0)} = A$

for $k = 1, 2, \ldots$

$\quad Q^{(k)}R^{(k)} = A^{(k-1)}$

$\quad A^{(k)} = R^{(k)}Q^{(k)}$

Simultaneous Inverse Iteration $\Longleftrightarrow$ QR Algorithm

- Let $\underline{Q}^{(k)} = Q^{(1)}Q^{(2)}\cdots Q^{(k)}$ and $\underline{R}^{(k)} = R^{(k)}R^{(k-1)}\cdots R^{(1)}$. Then $A^k = \underline{Q}^{(k)}\underline{R}^{(k)}$.
- Inverting A^k, we have $A^{-k} = \left(\underline{R}^{(k)}\right)^{-1}\left(\underline{Q}^{(k)}\right)^{\mathsf{T}}$.
- Because A^{-k} is symmetric, $A^{-k} = \underline{Q}^{(k)}\left(\underline{R}^{(k)}\right)^{-\mathsf{T}}$.
- Using the "flipped" permutation matrix P, we can write the last expression as

$$A^{-k}P = \left[\underline{Q}^{(k)}P\right]\left[P\left(\underline{R}^{(k)}\right)^{-\mathsf{T}}P\right],$$

  which is the QR factorization of $A^{-k}P$.
- Therefore, simultaneous inverse iteration applied to $\hat{Q}^{(0)} = P$ is "equivalent" to the QR algorithm, in that it produces

$$\hat{Q}^{(k)} = \underline{Q}^{(k)}P \quad \text{and} \quad \hat{R}^{(k)}\hat{R}^{(k-1)}\cdots\hat{R}^{(1)} = P\left(\underline{R}^{(k)}\right)^{-\mathsf{T}}P.$$

- Question: How to obtain $A^{(k)}$ in simultaneous inverse iteration?
  Answer: $A^{(k)} = \left(\underline{Q}^{(k)}\right)^{\mathsf{T}}A\underline{Q}^{(k)} = P\left(\hat{Q}^{(k)}\right)^{\mathsf{T}}A\hat{Q}^{(k)}P$.

QR Algorithm with Shifts

- Similar to inverse iteration, we can introduce shifts $\mu^{(k)}$ to accelerate convergence.

QR Algorithm with Shifts

$A^{(0)} = A$

for $k = 1, 2, \ldots$

Pick a shift $\mu^{(k)}$

$Q^{(k)}R^{(k)} = A^{(k-1)} - \mu^{(k)}I$

$A^{(k)} = R^{(k)}Q^{(k)} + \mu^{(k)}I$

Properties of QR Algorithm with Shift

- From $Q^{(k)}R^{(k)} = A^{(k-1)} - \mu^{(k)}I$ and $A^{(k)} = R^{(k)}Q^{(k)} + \mu^{(k)}I$, we have

$$A^{(k)} = \left(Q^{(k)}\right)^{\mathsf{T}} A^{(k-1)} Q^{(k)}.$$

- Then by induction, $A^{(k)} = \left(\underline{Q}^{(k)}\right)^{\mathsf{T}} A \underline{Q}^{(k)}$.
- However, instead of $A^k = \underline{Q}^{(k)}\underline{R}^{(k)}$, we now have

$$\left(A - \mu^{(k)}I\right)\left(A - \mu^{(k-1)}I\right)\cdots\left(A - \mu^{(1)}I\right) = \underline{Q}^{(k)}\underline{R}^{(k)},$$

  which can be shown by induction.
- In other words, $\underline{Q}^{(k)}$ is the orthogonalization of $\prod_{j=k}^{1}(A - \mu^{(j)}I)$.
- If $\mu^{(k)}$ are good estimates of eigenvalues, then the last column of $\underline{Q}^{(k)}$ converges to the corresponding eigenvector.

Choosing $\mu^{(k)}$: Rayleigh Quotient Shift

- A natural choice of $\mu^{(k)}$ is the Rayleigh quotient for the last column of $\underline{Q}^{(k)}$, given by

$$\mu^{(k)} = r(q_n^{(k)}) = \left(q_n^{(k)}\right)^{\mathsf{T}} A q_n^{(k)}.$$

- As in Rayleigh quotient iteration, the last column $q_n^{(k)}$ converges cubically.
- This Rayleigh quotient appears as the (n, n) entry of $A^{(k)}$ since $A^{(k)} = \left(\underline{Q}^{(k)}\right)^{\mathsf{T}} A \underline{Q}^{(k)}$.
- The Rayleigh quotient shift corresponds to setting $\mu^{(k)} = A_{nn}^{(k)}$.

Choosing $\mu^{(k)}$: Wilkinson Shift

- The QR algorithm with the Rayleigh quotient shift might fail sometimes, e.g., $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, for which $A^{(k)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and μ is always 0.
- Wilkinson breaks the symmetry by considering the lower-rightmost 2×2 submatrix of $A^{(k)}$ given by $B = \begin{bmatrix} a_{n-1} & b_{n-1} \\ b_{n-1} & a_n \end{bmatrix}$.
- We choose the eigenvalue of B closer to a_n, with arbitrary tie-breaking as

$$\mu = a_n - \text{sign}(\delta) b_{n-1}^2 / \left(|\delta| + \sqrt{\delta^2 + b_{n-1}^2} \right),$$

  where $\delta = (a_{n-1} - a_n)/2$. If $\delta = 0$, set $\text{sign}(\delta)$ to 1 (or -1).
- The QR algorithm always converges with this shift; quadratically in the worst case, and cubically in general.

“Practical” QR Algorithm

- The “practical” QR algorithm involves two additional components:
  - Tridiagonalization of A at the beginning. The tridiagonal structure is preserved by $A^{(k)}$.
  - Deflation of A into submatrices when $A^{(k)}$ is separable.

“Practical” QR Algorithm

$\left(Q^{(0)}\right)^{\mathsf{T}} A^{(0)} Q^{(0)} = A$ {tridiagonalization of A}

for $k = 1, 2, \ldots$

Pick a shift $\mu^{(k)}$

$Q^{(k)} R^{(k)} = A^{(k-1)} - \mu^{(k)} I$

$A^{(k)} = R^{(k)} Q^{(k)} + \mu^{(k)} I$

If any off-diagonal element $a_{j,j+1}^{(k)}$ is sufficiently close to zero

set $a_{j,j+1} = a_{j+1,j} = 0$ to obtain $\begin{bmatrix} A_1 & \\ & A_2 \end{bmatrix} = A^{(k)}$ and apply the QR algorithm to A_1 and A_2.

Stability and Accuracy

Theorem

The QR algorithm is backward stable, satisfying

$$\tilde{Q}\tilde{\Lambda}\tilde{Q}^{\mathsf{T}} = A + \delta A, \quad \|\delta A\|/\|A\| = \mathcal{O}(\epsilon_{\text{machine}}),$$

where $\tilde{\Lambda}$ is the computed Λ and $\tilde{Q}$ is an exactly orthogonal matrix.

- Its combination with Hessenberg reduction is also backward stable.
- Furthermore, eigenvalues are always well conditioned for **normal** matrices in that $|\tilde{\lambda}_j - \lambda_j| \leq \|\delta A\|_2$ and hence

$$|\tilde{\lambda}_j - \lambda_j|/\|A\| = \mathcal{O}(\epsilon_{\text{machine}}),$$

  where $\tilde{\lambda}_j$ are the computed eigenvalues.
- However, the sensitivity of eigenvectors depends on distances between adjacent eigenvalues, so the error in eigenvectors may be arbitrarily large.